Predicting Flight Delays and Their Causes Across Major U.S. Airports

Exploratory Data Analysis
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GitHub repository: https://github.com/USERNAME/flight-delay-project

Replace USERNAME with the public repository URL before submission.

Summary of Research Questions

1. Which factors, including airport, airline, scheduled time, and reported delay cause, are most associated with longer flight delays?

2. Which airports in the available extract have the longest average departure and arrival delays?

3. Does the reported weather-delay category identify flights that are more likely to have departure delays longer than 30 minutes?

Motivation

Flight delays affect millions of travelers and can result from more than weather alone. Airport congestion, airline operations, late arriving aircraft, and scheduled departure time can all contribute. Because I am interested in aviation and am pursuing pilot training, I want to understand which factors are most strongly connected to delays. This analysis provides practice working with real aviation records while also producing findings that could help travelers make more informed scheduling decisions.

Dataset

The EDA uses a 77-row exploratory extract from a public 2015 U.S. flight-delay dataset derived from Bureau of Transportation Statistics on-time performance records. Each row represents one flight record associated with a destination airport. The extract contains 17 original columns: date components, airline, origin and destination, scheduled departure, departure and arrival delay, distance, cancellation indicator, reported delay cause, and airport name/location attributes. Three engineered columns were added during analysis: departure hour, time-of-day category, and an indicator for a departure delay greater than 30 minutes.

The extract covers Seattle-Tacoma International (SEA), Dallas/Fort Worth International (DFW), and Chicago O'Hare International (ORD). Airport attributes are already joined to the flight records. The original proposal planned a separate NOAA weather merge; because the NOAA observations were not available in the submitted files, this EDA uses the existing Weather Delay category rather than measured precipitation, visibility, temperature, or wind. Therefore, the multiple-dataset goal is only partially completed at this stage.

Method

The Python script loads the CSV with pandas, creates a Boolean DELAY_OVER_30 variable, converts scheduled departure times into hours, and groups hours into overnight, morning, afternoon, evening, and late-night categories. Missing values are measured with DataFrame.isna().sum(). Quantitative variables are summarized with DataFrame.describe(), while categorical variables are summarized with value_counts(). The analysis then groups records by destination airport and reported delay cause. Three matplotlib bar charts compare average airport delay, delay-cause counts, and the percentage delayed over 30 minutes by time of day.

EDA Results

Size, Rows, and Columns

The raw extract has 77 rows and 17 columns. After feature engineering, the analysis table has 20 columns. A row represents a 2015 flight record, and columns describe the date, airline, route, timing, delay outcomes, distance, reported cause, and destination-airport attributes.

Missing Data

Pythonic missing-value checks found two missing ARRIVAL_DELAY values, equal to 2.60% of the records. Every other analyzed column had zero missing values. The two missing arrival delays are retained for analyses that do not use arrival delay and are automatically excluded by pandas when calculating the arrival-delay mean. A later model should either exclude those rows for an arrival-delay outcome or investigate whether the values correspond to cancellations or incomplete reporting.

Variables of Interest and Summaries

The primary quantitative variables are DEPARTURE_DELAY, ARRIVAL_DELAY, and DISTANCE. Airport code, airline, time of day, and reported delay cause are categorical variables. DEPARTURE_DELAY and ARRIVAL_DELAY directly answer which airports and scheduling groups experience the longest delays. DELAY_OVER_30 is the proposed classification outcome. CANCELLATION_REASON is used as the available proxy for operational or weather-related cause.

Variable	Mean	SD	Min	Q1	Median	Q3	Max
Departure delay	40.47	25.76	16	23	34	41	100
Arrival delay	22.79	32.04	0	5	13	23.5	171
Distance	581.22	503.59	67	247	491	802	3711

Categorical counts: Carrier Delay = 44 (57.1%), National Air System Delay = 16 (20.8%), Weather Delay = 15 (19.5%), and Late Aircraft Delay = 2 (2.6%). Airline counts were MQ = 42, AA = 13, OO = 13, AS = 4, EV = 3, DL = 1, and NK = 1. These counts show that the extract is not balanced across airlines, so airline comparisons would not yet be reliable.

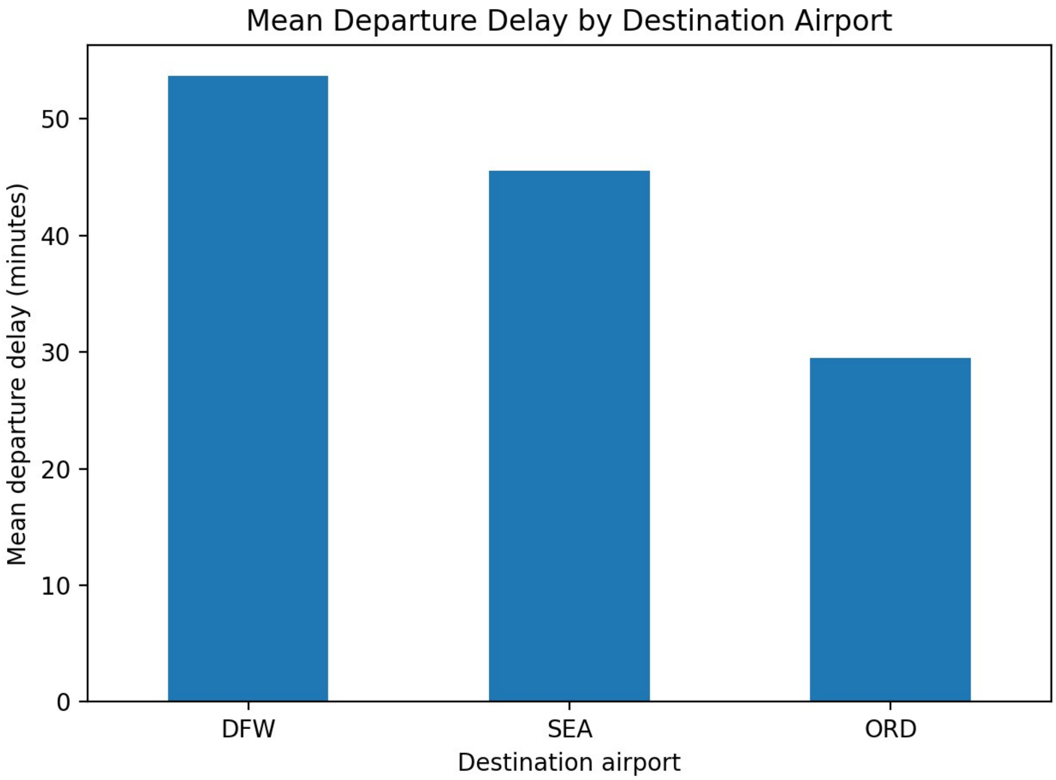

Figure 1. DFW has the highest mean departure delay in this extract, followed by SEA and ORD.

This bar chart compares the mean departure delay for the three destination airports. A bar chart is appropriate because airport is categorical and the outcome is a group mean. DFW had the highest mean departure delay at 53.68 minutes, SEA averaged 45.53 minutes, and ORD averaged 29.49 minutes. The percentage delayed more than 30 minutes was 72.0% at DFW, 80.0% at SEA, and 48.6% at ORD. These values describe only this small extract and should not be interpreted as rankings for all 2015 flights.

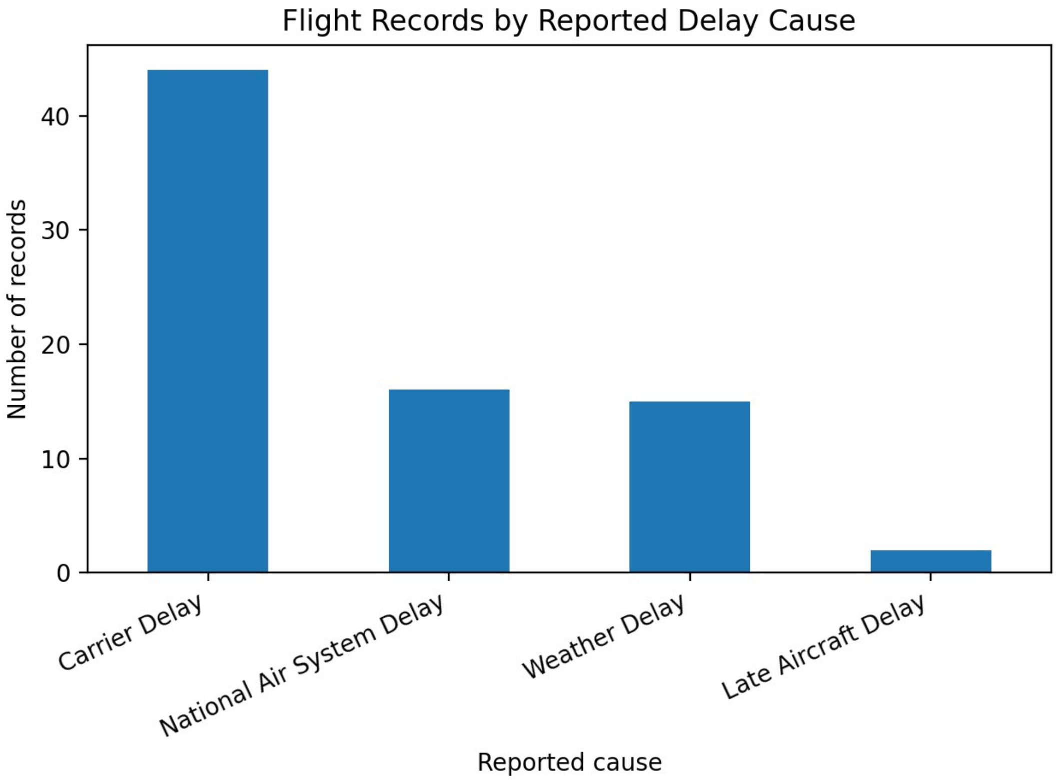

Figure 2. Carrier-related records are the most common, while weather accounts for about one-fifth of the extract.

The second plot displays the frequency of each reported delay-cause category. It shows that carrier delay is most common, with 44 records, followed by National Air System delay with 16 and weather delay with 15. This supports the motivation that delays are not caused by weather alone. However, these labels are not direct weather measurements and may also reflect the way the source extract was sampled.

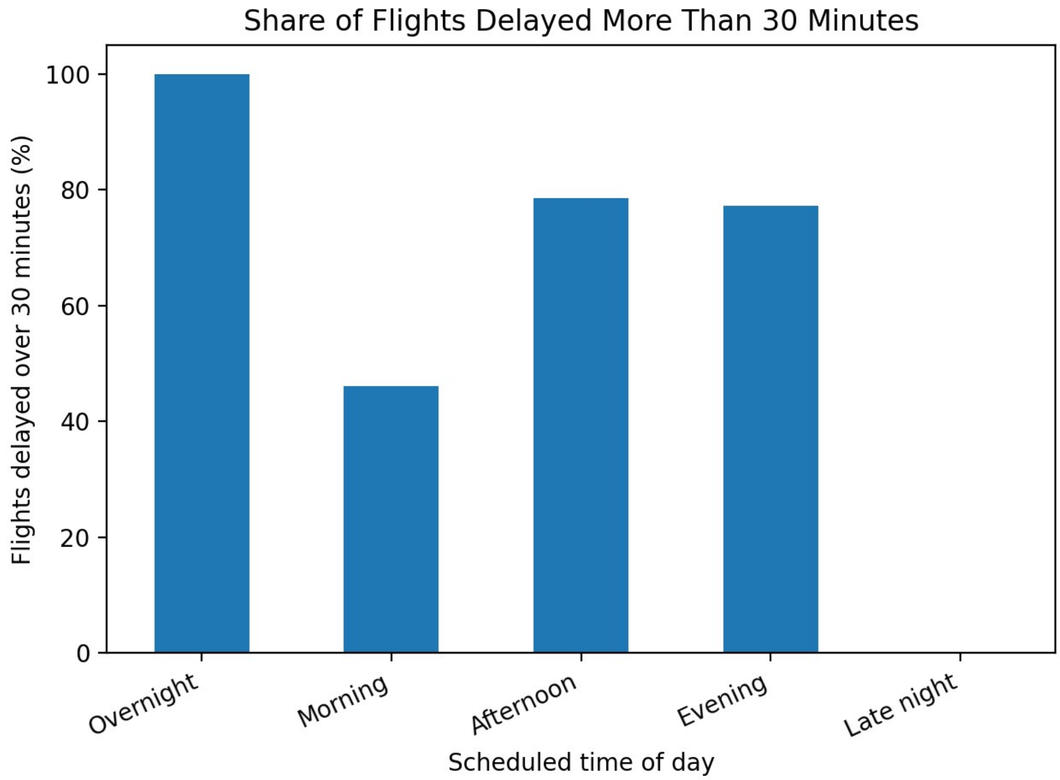

Figure 3. Afternoon and evening flights have higher over-30-minute delay shares than morning flights in the extract.

The third plot compares the percentage of records with departure delay over 30 minutes across scheduled time-of-day groups. Afternoon flights had a 78.6% rate and evening flights had a 77.3% rate, compared with 46.2% in the morning. The overnight category contains only two records, so its 100% rate is not stable. The pattern suggests that delay accumulation later in the day is worth testing with a larger and more representative dataset.

Challenge Goals Update

The multiple-datasets goal has been scaled back for the EDA. Airport information is present as prejoined attributes, but separate NOAA weather observations have not yet been merged. The next step is to obtain hourly or daily NOAA observations and join them by airport and date/time. The machine-learning goal remains planned: logistic regression, random forest, and gradient boosting can be compared after the full dataset is assembled. The target will be DELAY_OVER_30, and evaluation will include accuracy, precision, recall, F1 score, and a confusion matrix.

Work Plan Evaluation

The original plan underestimated the time needed to locate compatible datasets and create reliable join keys. Cleaning and basic visualization were close to the expected effort, but weather integration took longer because station identifiers and observation times do not directly match flight airport codes and scheduled times. Remaining work includes obtaining the full BTS records, mapping NOAA stations to airports, validating the merge, expanding the EDA, and training models. I estimate approximately 3 hours for data acquisition and merging, 2 hours for expanded testing and EDA, 3 hours for modeling, and 2 hours for final report revision.

Testing

The testing program contains five pytest tests. It checks the expected row count and engineered columns, verifies that missing-value totals match a direct pandas calculation, compares summary minimum and maximum values with independent calculations, confirms that airport grouping accounts for every record exactly once, and tests the loader on a controlled two-row CSV with known delay classifications. The full test suite passed. The EDA script was also run from its main method to regenerate every table and figure. Flake8 was not installed in the execution environment, so the files were manually formatted to follow standard line length, naming, documentation, and main-method conventions; the student should run flake8 locally before submission.

Collaboration

Resources consulted include the course assignment instructions, pandas and matplotlib documentation, the Bureau of Transportation Statistics description of on-time performance data, and the public 2015 flight-delay extract used for this EDA. No collaborators outside the course staff and publicly available documentation contributed analysis or code.

Data limitation: This submission is a complete reproducible EDA draft based on a small public extract. Conclusions are exploratory and should be updated after replacing flight_sample.csv with the full project datasets and rerunning eda.py.